

**Dr.K.V.SUBBAREDDY COLLEGE OF ENGINEERING
FOR WOMEN**

DUPADU, KURNOOL

**Department of Electronics & Communications Engineering
IV B.Tech-I Sem (E.C.E)**

VLSI
LAB
MANUAL
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**Dr.K.V.SUBBAREDDY COLLEGE OF ENGINEERING
FOR WOMEN**

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Department of Electronics & Communications Engineering

IV B.Tech-I Sem (E.C.E)

1. Logic Gates
2. 3 to 8 Decoder – 74138
3. 4 bit Comparator – 7485
4. 8x1 Multiplexer – 74151 and 2x4 Demultiplexer – 74155
5. D Flip-Flop -7474
6. Shift Register – 74957
7. Decade Counter – 7490

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IV B.Tech-I Sem (E.C.E)

LOGIC GATES

Exp No: 1

AIM: Simulate the programs of all Logic Gates using VHDL in Data flow, Behavioral and Structural models.

APPARATUS REQUIRED:

XILINX 9.2I /Active HDL4.2 software.

AND Gate

PROGRAM:

Data flow Model

```
library IEEE;
use IEEE.std_logic_1164.all;
entity and2_dat is
    port ( a: in STD_LOGIC;
           b: in STD_LOGIC;
           c: out STD_LOGIC);
end and2;
architecture and2_dat of and2_dat is
begin
    c<= a and b;
end and2_dat;
```

Behavioral Model

```
library IEEE;
use IEEE.std_logic_1164.all;
entity and2_beh is
    port ( a: in STD_LOGIC;
           b: in STD_LOGIC;
           c: out STD_LOGIC);
end and2_beh;

architecture and2_beh of and2_beh is
begin
    process(a,b)
    begin
        if(a='1' and b='1') then c<='1';
        else c<='0';
        end if;
    end process;
end and2_beh;
```

Structural Model

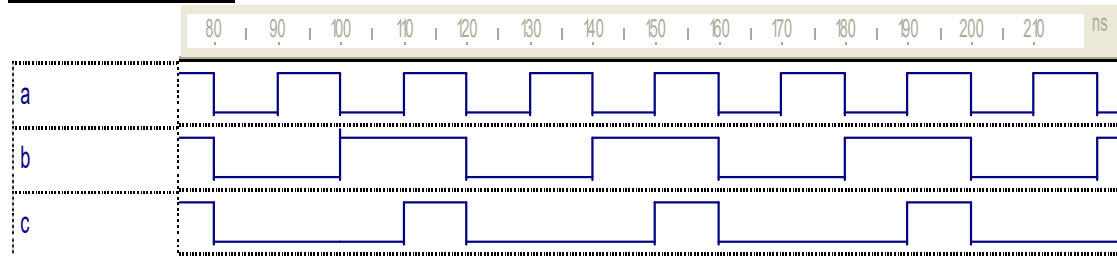
```
library IEEE;
use IEEE.std_logic_1164.all;
entity and2_str is
```

```

        port ( a: in STD_LOGIC;
              b: in STD_LOGIC;
              c: out STD_LOGIC);
end and2_str;
architecture and2_str of and2_str is
component and2_dat is
    port(a:in STD_LOGIC;
         b: in STD_LOGIC;
         c: out STD_LOGIC);
end component;
begin
a1: and2_dat port map(a,b,c);
end and2_str;
Test bench:
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
entity and_tb is
-- Port ( );
end and_tb;
architecture Behavioral of and_tb is
--Component name and entity's name must be same
--ports must be same
component and2_dat is
    Port (A,B:in std_logic;
          C: out std_logic );
end component;
--inputs
signal a: std_logic:= '0';
signal b: std_logic:= '0';
--outputs
signal c : std_logic;
begin
uut: and2_dat PORT MAP(a=>A,b=>B,c=>C);
--Stimulus Process
stim_proc:process
begin
a<='1';b<='0';
wait for 10ns;
a<='0';b<='1';
wait for 10ns;
a<='0';b<='0';
wait for 10ns;
a<='1';b<='1';
wait;
end process;
end Behavioral;

```

WAVEFORMS:



OR Gate

PROGRAM:

Data flow Model

```
library IEEE;
use IEEE.std_logic_1164.all;
entity or2_dat is
    port ( a: in STD_LOGIC;
           b: in STD_LOGIC;
           c: out STD_LOGIC);
end or2;

architecture or2_dat of or2_dat is
begin
    c<= a or b;
end or2_dat;
```

Behavioral Model

```
library IEEE;
use IEEE.std_logic_1164.all;
entity or2_beh is
    port ( a: in STD_LOGIC;
           b: in STD_LOGIC;
           c: out STD_LOGIC);
end or2_beh;
architecture or2_beh of or2_beh is
begin
    process(a,b)
    begin
        if(a='0' and b='0') then c<='0';
        else c<='1';
        end if;
    end process;
end or2_beh;
```

Structural Model

```
library IEEE;
use IEEE.std_logic_1164.all;
entity or2_str is
    port ( a: in STD_LOGIC;
           b: in STD_LOGIC;
           c: out STD_LOGIC);
end or2_str;
```

```

architecture or2_str of or2_str is
component or2_dat is
    port(a:in STD_LOGIC;
          b: in STD_LOGIC;
          c: out STD_LOGIC);
end component;
begin
a1: or2_dat port map(a,b,c);
end or2_str;

```

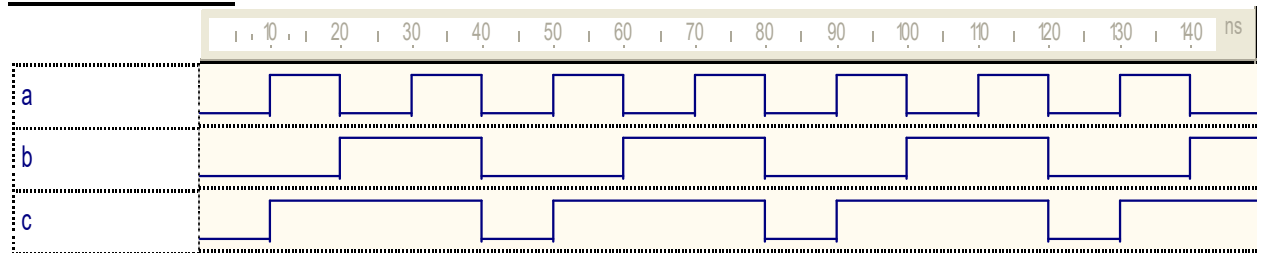
TESTBENCH:

```

library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
entity or_tb is
-- Port ();
end or_tb;
architecture Behavioral of or_tb is
--Component name and entity's name must be same
--ports must be same
component or2_dat is
    Port (A,B:in std_logic;
          C: out std_logic );
end component;
--inputs
signal a: std_logic:= '0';
signal b: std_logic:= '0';
--outputs
signal c : std_logic;
begin
uut: or2_dat PORT MAP(a=>A,b=>B,c=>C);
--Stimulus Process
stim_proc:process
begin
wait for 10ns;
a<='1';b<='0';
wait for 10ns;
a<='0';b<='1';
wait for 10ns;
a<='0';b<='0';
wait for 10ns;
a<='1';b<='1';
wait for 10ns;
end process;
end Behavioral;

```

WAVEFORMS:



NOT Gate

PROGRAM:

Data flow Model

```
library IEEE;
use IEEE.std_logic_1164.all;
entity not2_dat is
    port ( a: in STD_LOGIC;
           c: out STD_LOGIC);
end not2;
architecture not2_dat of not2_dat is
begin
    c<= not a ;
end not2_dat;
```

Behavioral Model

```
library IEEE;
use IEEE.std_logic_1164.all;
entity not2_beh is
    port ( a: in STD_LOGIC;
           c: out STD_LOGIC);
end not2_beh;
architecture not2_beh of not2_beh is
begin
    process(a)
    begin
        if(a='0' ) then c<='1';
        else c<='0';
        end if;
    end process;
end not2_beh;
```

Structural Model

```
library IEEE;
use IEEE.std_logic_1164.all;

entity nor2_str is
    port ( a: in STD_LOGIC;
           c: out STD_LOGIC);
end nor2_str;
architecture not2_str of not2_str is
```

```

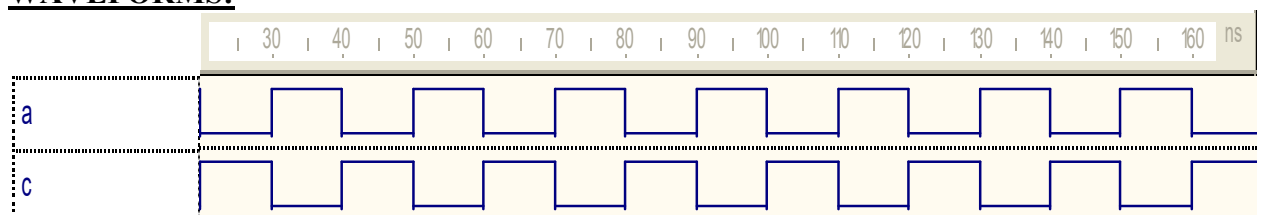
component not2_dat is
    port(a:in STD_LOGIC;
          c: out STD_LOGIC);
end component;
begin
a1: not2_dat port map(a,c);
end or2_str;
Test bench
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
entity not2_tb is
    -- Port ( );
end not2_tb;

architecture Behavioral of not2_tb is
    --Component name and entity's name must be same
    --ports must be same
    component not2_dat is
        Port (A:in std_logic;
              C: out std_logic );
    end component;
    --inputs
    signal a: std_logic:= '0';
    --outputs
    signal c : std_logic;

begin
    uut: not2_dat PORT MAP(a=>A,c=>C);
    --Stimulus Process
    stim_proc:process
    begin
        wait for 10ns;
        a<='1';
        wait for 10ns;
        a<='0';
        wait for 10ns;
        a<='0';
        wait for 10ns;
        a<='1';
        wait for 10ns;
    end process;
end Behavioral;

```

WAVEFORMS:



NAND Gate

PROGRAM:

Data flow Model

```
library IEEE;
use IEEE.std_logic_1164.all;
entity nand2_dat is
    port ( a: in STD_LOGIC;
           b: in STD_LOGIC;
           c: out STD_LOGIC);
end nand2;
architecture nand2_dat of nand2_dat is
begin
    c<= a nand b;
end nand2_dat;
```

Behavioral Model

```
library IEEE;
use IEEE.std_logic_1164.all;
entity nand2_beh is
    port ( a: in STD_LOGIC;
           b: in STD_LOGIC;
           c: out STD_LOGIC);
end nand2_beh;

architecture nand2_beh of nand2_beh is
begin
    process(a,b)
    begin
        if(a='1' and b='1') then c<='0';
        else c<='1';
        end if;
    end process;
end nand2_beh;
```

Structural Model

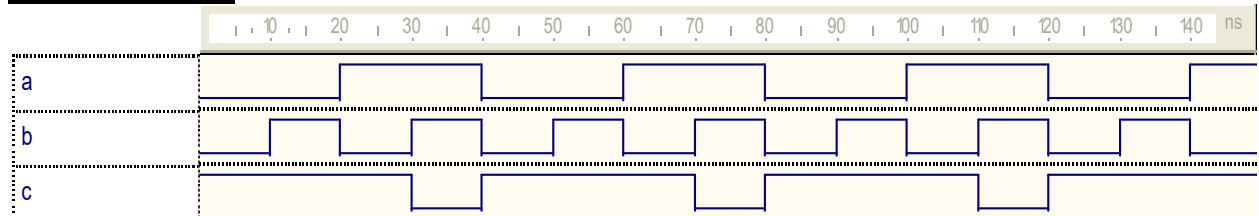
```
library IEEE;
use IEEE.std_logic_1164.all;
entity nand2_str is
    port ( a: in STD_LOGIC;
           b: in STD_LOGIC;
           c: out STD_LOGIC);
end nand2_str;
architecture nand2_str of nand2_str is
    component and2_dat is
        port(a:in STD_LOGIC;
             b: in STD_LOGIC;
             c: out STD_LOGIC);
    end component;
    component not2_dat is
        port( a: in STD_LOGIC;
              c: out STD_LOGIC);
```

```

end component;
signal temp:std_logic;
begin
    a1:and2_dat port map(a,b,temp);
    n1:not2_dat port map(temp,c);
end nand2_str;

```

WAVEFORMS:



NOR Gate

PROGRAM:

Data flow Model

```

library IEEE;
use IEEE.std_logic_1164.all;
entity nor2_dat is
    port ( a: in STD_LOGIC;
           b: in STD_LOGIC;
           c: out STD_LOGIC);
end nor2;
architecture nor2_dat of nor2_dat is
begin
    c<= a nor b;
end nor2_dat;

```

Behavioral Model

```

library IEEE;
use IEEE.std_logic_1164.all;
entity nor2_beh is
    port ( a: in STD_LOGIC;
           b: in STD_LOGIC;
           c: out STD_LOGIC);
end nor2_beh;
architecture nor2_beh of nor2_beh is
begin
    process(a,b)
    begin
        if(a='0' and b='0') then c<='1';
        else c<='0';
        end if;
    end process;
end nor2_beh;

```

Structural Model

```

library IEEE;

```

```

use IEEE.std_logic_1164.all;
entity nor2_str is
    port ( a: in STD_LOGIC;
           b: in STD_LOGIC;
           c: out STD_LOGIC);
end nor2_str;
architecture nor2_str of nor2_str is
    component or2_dat is
        port(a:in STD_LOGIC;
             b: in STD_LOGIC;
             c: out STD_LOGIC);
    end component;
    component not2_dat is
        port( a: in STD_LOGIC;
              c: out STD_LOGIC);
    end component;
    signal temp: std_logic;
begin
    a1:or2_dat port map(a,b,temp);
    n1:not2_dat port map(temp,c);
end nor2_str;

```

TestBench

```

library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
entity nor2_tb is
    -- Port ( );
end nor2_tb;

architecture Behavioral of nor2_tb is
    --Component name and entity's name must be same
    --ports must be same
    component nor2_dat is
        Port (A,B:in std_logic;
              C: out std_logic );
    end component;
    --inputs
    signal a: std_logic:= '0';
    signal b: std_logic:= '0';
    --outputs
    signal c : std_logic;

begin
    uut: nor2_dat PORT MAP(a=>A,b=>B,c=>C);
    --Stimulus Process
    stim_proc:process
    begin
        wait for 10ns;
        a<='1';b<='0';
        wait for 10ns;
    end process;
end Behavioral;

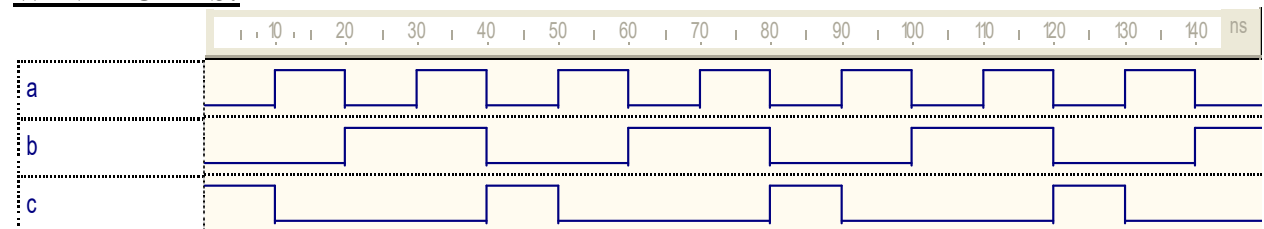
```

```

a<='0';b<='1';
wait for 10ns;
a<='0';b<='0';
wait for 10ns;
a<='1';b<='1';
wait for 10ns;
end process;
end Behavioral;

```

WAVEFORMS:



XOR Gate

PROGRAM:

Data flow Model

```

library IEEE;
use IEEE.std_logic_1164.all;
entity xor2_dat is
    port ( a: in STD_LOGIC;
           b: in STD_LOGIC;
           c: out STD_LOGIC);
end xor2;
architecture xor2_dat of xor2_dat is
begin
    c<= a xor b;
end xor2_dat;

```

Behavioral Model

```

library IEEE;
use IEEE.std_logic_1164.all;
entity xor2_beh is
    port ( a: in STD_LOGIC;
           b: in STD_LOGIC;
           c: out STD_LOGIC);
end xor2_beh;
architecture xor2_beh of xor2_beh is
begin
    process(a,b)
    begin
        if(a= b) then c<='0';
        else c<='1';
        end if;
    end process;
end xor2_beh;

```

Structural Model

```

library IEEE;

```

```

use IEEE.std_logic_1164.all;
entity xor2_str is
    port ( a: in STD_LOGIC;
           b: in STD_LOGIC;
           c: out STD_LOGIC);
end xor2_str;
architecture xor2_str of xor2_str is
    component or2_dat is
        port(a:in STD_LOGIC;
             b: in STD_LOGIC;
             c: out STD_LOGIC);
    end component;
    component not2_dat is
        port( a: in STD_LOGIC;
              c: out STD_LOGIC);
    end component;
    component and2_dat is
        port ( a: in STD_LOGIC;
               b: in STD_LOGIC;
               c: out STD_LOGIC);
    end component;
    signal t1,t2,t3,t4:std_logic;
    begin
        n1:not2_dat port map(a,t1);
        a1:and2_dat port map(t1,b,t2);
        n2:not2_dat port map(b,t3);
        a2:and2_dat port map(t3,a,t4);
        r1:or2_dat port map(t2,t4,c);
        end xor2_str;

```

TESTBENCH:

```

library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
entity xor2_tb is
    -- Port ( );
end xor2_tb;

architecture Behavioral of xor2_tb is
    --Component name and entity's name must be same
    --ports must be same
    component xor2_dat is
        Port (A,B:in std_logic;
              C: out std_logic );
    end component;
    --inputs
    signal a: std_logic:= '0';
    signal b: std_logic:= '0';
    --outputs
    signal c : std_logic;

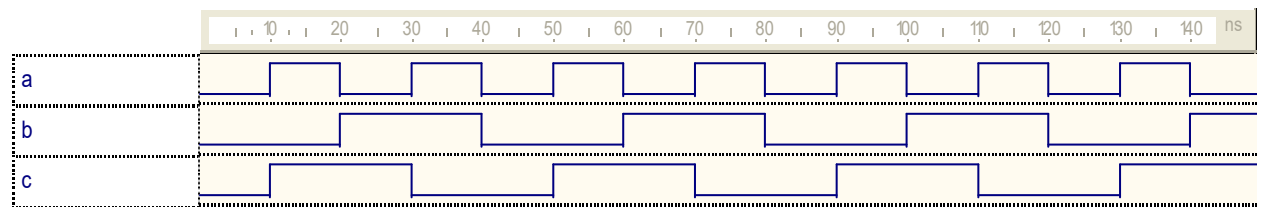
```

```

begin
uut: xor2_dat PORT MAP(a=>A,b=>B,c=>C);
--Stimulus Process
stim_proc:process
begin
wait for 10ns;
a<='1';b<='0';
wait for 10ns;
a<='0';b<='1';
wait for 10ns;
a<='0';b<='0';
wait for 10ns;
a<='1';b<='1';
wait for 10ns;
end process;
end Behavioral;

```

WAVEFORMS:



RESULT: All models of Logic Gates are simulated and waveforms are observed.

Exp No:2

AIM: Simulate the programs of 3-to-8 Decoder using VHDL in Data flow, Behavioral and Structural models.

APPARATUS REQUIRED:

XILINX 9.2I /Active HDL4.2 software.

PROGRAM:

Data flow Model

```
library IEEE;
use IEEE.std_logic_1164.all;
entity decdr3x8_dat is
    port ( a: in STD_LOGIC;
           b: in STD_LOGIC;
           c: in STD_LOGIC;
           y0: out STD_LOGIC;
           y1: out STD_LOGIC;
           y2: out STD_LOGIC;
           y3: out STD_LOGIC;
           y4: out STD_LOGIC;
           y5: out STD_LOGIC;
           y6: out STD_LOGIC;
           y7: out STD_LOGIC);
end decdr3x8_dat;
architecture decdr3x8_dat of decdr3x8_dat is
begin
    y0<=(not a) and (not b) and (not c);
    y1<=(not a) and (not b) and c;
    y2<=(not a) and b and (not c);
    y3<=(not a) and (b and c);
    y4<= a and (not b) and (not c);
    y5<= (a and c) and (not b);
    y6<= (a and b) and (not c);
    y7<= (a and b) and c;
end decdr3x8_dat;
```

Behavioral Model

```
library IEEE;
use IEEE.std_logic_1164.all;
entity decdr3x8_beh is
    port (a: in STD_LOGIC_VECTOR (2 downto 0);
          en: in STD_LOGIC;
          y: out STD_LOGIC_VECTOR (7 downto 0));
end decdr3x8_beh;
architecture decdr3x8_beh of decdr3x8_beh is
```

```

begin
    process(en,a)
    begin
        if(en='1')then
            case a is
                when "000"=>y<="00000001";
                when "001"=>y<="00000010";
                when "010"=>y<="00000100";
                when "011"=>y<="00001000";
                when "100"=>y<="00010000";
                when "101"=>y<="00100000";
                when "110"=>y<="01000000";
                when "111"=>y<="10000000";
                when others=>y<="00000000";
            end case;
            else y<="00000000";
            end if;
        end process;
    end decdr3x8_beh;

```

Structural Model

--Program for 3 input AND Gate in data flow Model

```

library IEEE;
use IEEE.std_logic_1164.all;
entity and3_dat is
    port (a: in STD_LOGIC;
          b: in STD_LOGIC;
          c: in STD_LOGIC;
          p: out STD_LOGIC);
end and3_dat;
architecture and3_dat of and3_dat is
begin
    p<=a and b and c;
end and3_dat;

```

--Program for NOT Gate in Behavioral Model

```

library IEEE;
use IEEE.std_logic_1164.all;
entity not2_beh is
    port (a: in STD_LOGIC;
          c: out STD_LOGIC);
end not2_beh;
architecture not2_beh of not2_beh is
begin

```



```

process(a)
begin
    if(a='0') then c<='1';
        else c<='0';
    end if;
end process;
end not2_beh;

--Program for 3-to-8 Decoder in Structural Model
library IEEE;
use IEEE.std_logic_1164.all;
entity decdr3x8_str is
    port ( x: in STD_LOGIC;
           y: in STD_LOGIC;
           z: in STD_LOGIC;
           a0: out STD_LOGIC;
           a1: out STD_LOGIC;
           a2: out STD_LOGIC;
           a3: out STD_LOGIC;
           a4: out STD_LOGIC;
           a5: out STD_LOGIC;
           a6: out STD_LOGIC;
           a7: out STD_LOGIC);
end decdr3x8_str;
architecture decdr3x8_str of decdr3x8_str is
    component not2 is
        port( a:in STD_LOGIC;
              c:out STD_LOGIC);
    end component;
    component and3_dat is
        port( a,b,c: in STD_LOGIC;
              p: out STD_LOGIC);
    end component;
    signal t1,t3,t2: STD_LOGIC;
    begin
        n1: not2 port map(x,t1);
        n2: not2 port map(y,t2);
        n3: not2 port map(z,t3);
        d1: and3_dat port map(t1,t2,t3,a0);
        d2: and3_dat port map(t1,t2,z,a1);
        d3: and3_dat port map(t1,y,t3,a2);
        d4: and3_dat port map(t1,y,z,a3);
        d5: and3_dat port map(x,t2,t3,a4);

```

```

        d6:and3_dat port map(x,t2,z,a5);
        d7:and3_dat port map(x,y,t3,a6);
        d8:and3_dat port map(x,y,z,a7);
end decdr3x8_str;
TEST BENCH:
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;
USE ieee.std_logic_unsigned.all;
USE ieee.numeric_std.ALL;
ENTITY tb_kp328_vhd IS
END tb_kp328_vhd;
ARCHITECTURE behavior OF tb_kp328_vhd IS
    -- Component Declaration for the Unit Under Test (UUT)
    COMPONENT kp3t8de
    PORT(
        a : IN std_logic;
        b : IN std_logic;
        c : IN std_logic;
        y0 : OUT std_logic;
        y1 : OUT std_logic;
        y2 : OUT std_logic;
        y3 : OUT std_logic;
        y4 : OUT std_logic;
        y5 : OUT std_logic;
        y6 : OUT std_logic;
        y7 : OUT std_logic
    );
    END COMPONENT;

    --Inputs
    SIGNAL a : std_logic := '0';
    SIGNAL b : std_logic := '0';
    SIGNAL c : std_logic := '0';

    --Outputs
    SIGNAL y0 : std_logic;
    SIGNAL y1 : std_logic;
    SIGNAL y2 : std_logic;
    SIGNAL y3 : std_logic;
    SIGNAL y4 : std_logic;
    SIGNAL y5 : std_logic;
    SIGNAL y6 : std_logic;
    SIGNAL y7 : std_logic;

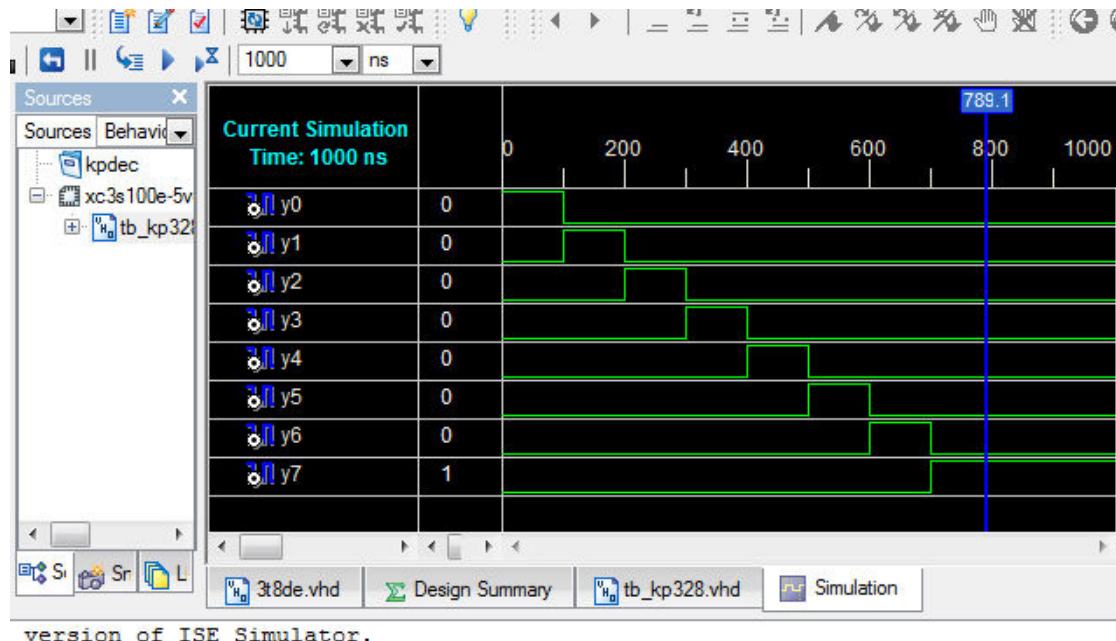
```

```

BEGIN
    -- Instantiate the Unit Under Test (UUT)
    uut: kp3t8de PORT MAP(
        a => a,
        b => b,
        c => c,
        y0 => y0,
        y1 => y1,
        y2 => y2,
        y3 => y3,
        y4 => y4,
        y5 => y5,
        y6 => y6,
        y7 => y7
    );

    tb : PROCESS
    BEGIN
        -- Wait 100 ns for global reset to finish
        a<='0'; b<='0'; c<='0';
        wait for 100 ns;
        a<='0'; b<='0'; c<='1';
        wait for 100 ns;
        a<='0'; b<='1'; c<='0';
        wait for 100 ns;
        a<='0'; b<='1'; c<='1';
        wait for 100 ns;
        a<='1'; b<='0'; c<='0';
        wait for 100 ns;
        a<='1'; b<='0'; c<='1';
        wait for 100 ns;
        a<='1'; b<='1'; c<='0';
        wait for 100 ns;
        a<='1'; b<='1'; c<='1';
        -- Place stimulus here
        wait; -- will wait forever
    END PROCESS;
END;
WAVEFORMS:

```



RESULT: All models of 3-to-8 Decoder are simulated and waveforms are observed.

3. 4 BIT COMPARATOR

Date:

Exp No: 3

AIM: Simulate the programs of 4 Bit Comparator using VHDL in Data flow, Behavioral and Structural models.

APPARATUS REQUIRED:

XILINX 9.2I /Active HDL4.2 software.

PROGRAM:

Data flow Model

```
library IEEE;
use IEEE.std_logic_1164.all;
entity comp4_dat is
    port ( a: in STD_LOGIC_VECTOR (3 downto 0);
          b: in STD_LOGIC_VECTOR (3 downto 0);
          agtb: out STD_LOGIC;
          aeqb: out STD_LOGIC;
          altb: out STD_LOGIC);
end comp4_dat;
architecture comp4_dat of comp4_dat is
begin
    altb<='1' when a<b else '0';
    aeqb<='1' when a=b else '0';
    agtb<='1' when a>b else '0';
```

```
end comp4_dat;
```

Behavioral Model

```
library IEEE;
use IEEE.std_logic_1164.all;
entity comp4_beh is
    port ( a: in STD_LOGIC_VECTOR (3 downto 0);
           b: in STD_LOGIC_VECTOR (3 downto 0);
           aeqb: out STD_LOGIC;
           agtb: out STD_LOGIC;
           altb: out STD_LOGIC);
end comp4_beh;

architecture comp4_beh of comp4_beh is
begin
    process(a,b)
    begin
        if(a=b) then aeqb<='1';
        else aeqb<='0';
            if(a>b) then agtb<='1';
            else agtb<='0';
                if(a<b) then altb<='1';
                else altb<='0';
                end if;
            end if;
        end if;
    end process;
end comp4_beh;
```

Structural Model

--Program for 2 input XNOR Gate in Data flow Model

```
library IEEE;
use IEEE.std_logic_1164.all;
entity xnor2_dat is
    port ( a: in STD_LOGIC;
           b: in STD_LOGIC;
           c: out STD_LOGIC);
end xnor2_dat;
architecture xnor2_dat of xnor2_dat is
begin
    c<=a xnor b;
end xnor2_dat;
```

--Program for NOT Gate in Behavioral Model

```
library IEEE;
use IEEE.std_logic_1164.all;
entity not2_beh is
    port ( a: in STD_LOGIC;
           c: out STD_LOGIC);
end not2_beh;
architecture not2_beh of not2_beh is
begin
    process(a)
    begin
        if(a='0') then c<='1';
        else c<='0';
        end if;
    end process;
end not2_beh;
```

--Program for 2 input AND Gate in Data flow Model

```
library IEEE;
use IEEE.std_logic_1164.all;
entity and2_dat is
    port ( a: in STD_LOGIC;
           b: in STD_LOGIC;
           c: out STD_LOGIC);
end and2_dat;
architecture and2_dat of and2_dat is
begin
    c<= a and b;
end and2_dat;
```

--Program for 3 input AND Gate in Data flow Model

```
library IEEE;
use IEEE.std_logic_1164.all;
entity and3_dat is
    port ( a: in STD_LOGIC;
           b: in STD_LOGIC;
           c: in STD_LOGIC;
           p: out STD_LOGIC);
end and3_dat;
architecture and3_dat of and3_dat is
begin
    p<=a and b and c;
end and3_dat;
```

--Program for 4 input AND Gate in Data flow Model

```
library IEEE;
use IEEE.std_logic_1164.all;
entity and4_dat is
    port ( a: in STD_LOGIC;
           b: in STD_LOGIC;
           c: in STD_LOGIC;
           d: in STD_LOGIC;
           p: out STD_LOGIC);
end and4_dat;
architecture and4_dat of and4_dat is
begin
    p<=a and b and c and d;
end and4_dat;
```

--Program for 5 input AND Gate in Data flow Model

```
library IEEE;
use IEEE.std_logic_1164.all;
entity and5_dat is
    port ( a: in STD_LOGIC;
           b: in STD_LOGIC;
           c: in STD_LOGIC;
           d: in STD_LOGIC;
           e: in STD_LOGIC;
           p: out STD_LOGIC);
end and5_dat;
architecture and5_dat of and5_dat is
begin
    p<=a and b and c and d and e;
end and5_dat;
```

--Program for 4 input OR Gate in Data flow Model

```
library IEEE;
use IEEE.std_logic_1164.all;
entity or4_dat is
    port ( a: in STD_LOGIC;
           b: in STD_LOGIC;
           c: in STD_LOGIC;
           d: in STD_LOGIC;
           p: out STD_LOGIC);
end or4_dat;
architecture or4_dat of or4_dat is
begin
```

```

        p<=a or b or c or d;
end or4_dat;

--Program for 4 Bit Comparator in Structural Model
library IEEE;
use IEEE.std_logic_1164.all;
entity comp4_str is
    port ( a: in STD_LOGIC_VECTOR (3 downto 0);
           b: in STD_LOGIC_VECTOR (3 downto 0);
           aeqb: out STD_LOGIC;
           agtb: out STD_LOGIC;
           altb: out STD_LOGIC);
end comp4_str;

architecture comp4_str of comp4_str is
    component xnor2_dat is
        port(a: in STD_LOGIC;
             b: in STD_LOGIC;
             c: out STD_LOGIC);
    end component;
    component not2_beh is
        port(a: in STD_LOGIC;
             c: out STD_LOGIC);
    end component;
    component and2_dat is
        port(a: in STD_LOGIC;
             b: in STD_LOGIC;
             c: out STD_LOGIC);
    end component;
    component and3_dat is
        port(a: in STD_LOGIC;
             b: in STD_LOGIC;
             c: in STD_LOGIC;
             p: out STD_LOGIC);
    end component;
    component and4_dat is
        port(a: in STD_LOGIC;
             b: in STD_LOGIC;
             c: in STD_LOGIC;
             d: in STD_LOGIC;
             p: out STD_LOGIC);
    end component;

```



```

component and5_dat is
    port(a: in STD_LOGIC;
          b: in STD_LOGIC;
          c: in STD_LOGIC;
          d: in STD_LOGIC;
          e: in STD_LOGIC;
          p: out STD_LOGIC);
end component;
component or4_dat is
    port(a: in STD_LOGIC;
          b: in STD_LOGIC;
          c: in STD_LOGIC;
          d: in STD_LOGIC;
          p: out STD_LOGIC);
end component;
signal t1,t2,t3,t4,s1,s2,s3,s4,s5,s6,s7,s8,s9,s10: STD_LOGIC;
begin
n1: not2_beh port map(b(0),t1);
n2: not2_beh port map(b(1),t2);
n3: not2_beh port map(b(2),t3);
n4: not2_beh port map(b(3),t4);
x1: xnor2_dat port map(a(0),b(0),s1);
x2: xnor2_dat port map(a(1),b(1),s2);
x3: xnor2_dat port map(a(2),b(2),s3);
x4: xnor2_dat port map(a(3),b(3),s4);
d1: and5_dat port map(a(0),t1,s2,s3,s4,s5);
d2: and4_dat port map(a(1),t2,s3,s4,s6);
d3: and3_dat port map(a(2),t3,s4,s7);
d4: and2_dat port map(a(3),t4,s8);
d5: and4_dat port map(s1,s2,s3,s4,s9);
d6: and4_dat port map(s1,s2,s3,s4,aeqb);
r1: or4_dat port map(s5,s6,s7,s8,s10);
r2: or4_dat port map(s5,s6,s7,s8,agtb);
x5: xnor2_dat port map(s9,s10,altb);
end comp4_str;

```

TEST BENCH:

```

LIBRARY ieee;
USE ieee.std_logic_1164.ALL;
USE ieee.std_logic_unsigned.all;
USE ieee.numeric_std.ALL;

```

```

ENTITY tb_kp4cmp_vhd IS
END tb_kp4cmp_vhd;

```

ARCHITECTURE behavior OF tb_kp4cmp_vhd IS

```
-- Component Declaration for the Unit Under Test (UUT)
COMPONENT kp4bcmp
PORT(
    a : IN std_logic_vector(3 downto 0);
    b : IN std_logic_vector(3 downto 0);
    agtb : OUT std_logic;
    aeqb : OUT std_logic;
    altb : OUT std_logic
);
END COMPONENT;

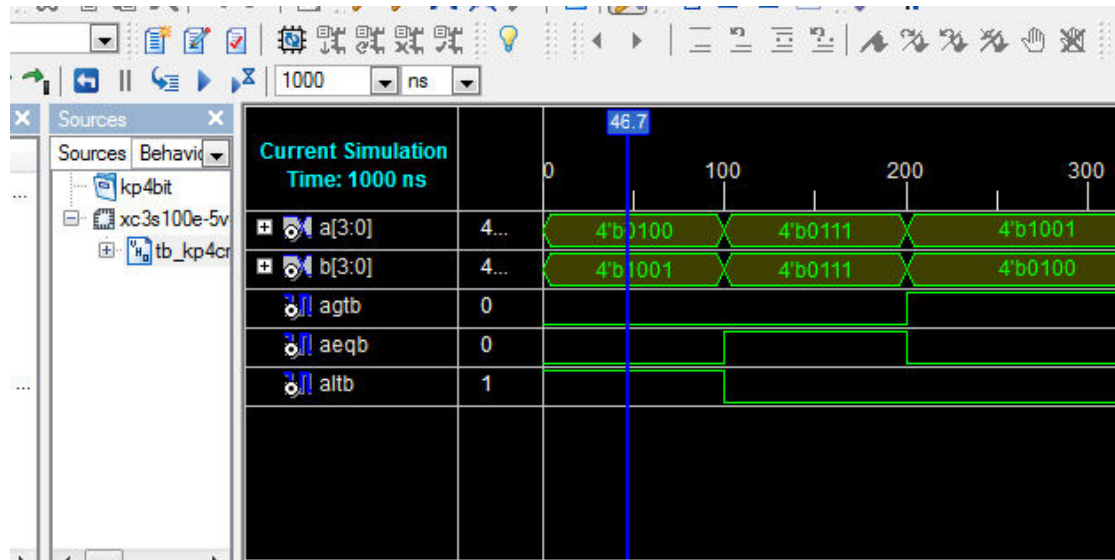
--Inputs
SIGNAL a : std_logic_vector(3 downto 0) := (others=>'0');
SIGNAL b : std_logic_vector(3 downto 0) := (others=>'0');
--Outputs
SIGNAL agtb : std_logic;
SIGNAL aeqb : std_logic;
SIGNAL altb : std_logic;
BEGIN
    -- Instantiate the Unit Under Test (UUT)
    uut: kp4bcmp PORT MAP(
        a => a,
        b => b,
        agtb => agtb,
        aeqb => aeqb,
        altb => altb
    );

    tb : PROCESS
    BEGIN

        -- Wait 100 ns for global reset to finish
        a<="0100"; b<="1001";
        wait for 100 ns;
        a<="0111"; b<="0111";
        wait for 100 ns;
        a<="1001"; b<="0100";
        -- Place stimulus here
        wait; -- will wait forever
    END PROCESS;
```

END;

WAVEFORMS:



RESULT: All models of 4 Bit Comparator are simulated and waveforms are observed.

4.1 8x1 MULTIPLEXER

Exp No: 4.1

AIM: Simulate the programs of 8X1 Multiplexer using VHDL in Data flow, Behavioral and Structural models.

APPARATUS REQUIRED:

XILINX 9.2I /Active HDL4.2 software.

PROGRAM:

Data flow Model

Data flow Model

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;
```

```
---- Uncomment the following library declaration if instantiating
---- any Xilinx primitives in this code.
--library UNISIM;
--use UNISIM.VComponents.all;
```

entity kpmux8x1 is

```
Port ( s : in STD_LOGIC_VECTOR (2 downto 0);
      d : in STD_LOGIC_VECTOR (7 downto 0);
      y : out STD_LOGIC);
```

```

end kpmux8x1;
architecture Behavioral of kpmux8x1 is
begin
  with s select
    y<= d(0) when "000",
    d(1) when "001",
    d(2)when "010",
    d(3)when "011",
    d(4) when "100",
    d(5)when "101",
    d(6)when "110",
    d(7) when others;
  --'U' when others;
end Behavioral;

```

Behavioral Model

```

library IEEE;
use IEEE.std_logic_1164.all;
entity mux8x1_beh is
  port ( a0: in STD_LOGIC;
        a1: in STD_LOGIC;
        a2: in STD_LOGIC;
        a3: in STD_LOGIC;
        a4: in STD_LOGIC;
        a5: in STD_LOGIC;
        a6: in STD_LOGIC;
        a7: in STD_LOGIC;
        en: in STD_LOGIC;
        s: in STD_LOGIC_VECTOR (2 downto 0);
        f: out STD_LOGIC);
end mux8x1_beh;
architecture mux8x1_beh of mux8x1_beh is
begin
  process(s)
  begin
    if(en='0') then
      case s is
        when "000"=>f<=a0;
        when "001"=>f<=a1;
        when "010"=>f<=a2;
        when "011"=>f<=a3;
        when "100"=>f<=a4;
        when "101"=>f<=a5;

```

```

        when "110"=>f<=a6;
        when "111"=>f<=a7;
        when others=>f<='U';
    end case;
    else f<='U';
end if;
end process;
end mux8x1_beh;

```

Structural Model

--Program for 4 input AND Gate in Data flow Model

```

library IEEE;
use IEEE.std_logic_1164.all;
entity and4_dat is
    port (a: in STD_LOGIC;
          b: in STD_LOGIC;
          c: in STD_LOGIC;
          d: in STD_LOGIC;
          p: out STD_LOGIC);
end and4_dat;
architecture and4_dat of and4_dat is
begin
    p<=a and b and c and d;
end and4_dat;

```

--Program for 8 input OR Gate in Data flow Model

```

library IEEE;
use IEEE.std_logic_1164.all;
entity or8_dat is
    port ( a: in STD_LOGIC;
          b: in STD_LOGIC;
          c: in STD_LOGIC;
          d: in STD_LOGIC;
          e: in STD_LOGIC;
          f: in STD_LOGIC;
          g: in STD_LOGIC;
          h: in STD_LOGIC;  p: out STD_LOGIC);
end or8_dat;
architecture or8_dat of or8_dat is
begin
    p<=a or b or c or d or e or f or g or h;
end or8_dat;

```

--Program for NOT Gate in Behavioral Model

```

library IEEE;
use IEEE.std_logic_1164.all;
entity not2_beh is
    port ( a: in STD_LOGIC;
           c: out STD_LOGIC);
end not2_beh;
architecture not2_beh of not2_beh is
begin
process(a)
begin
    If(a='0') then c<='1';
        else c<='0';
    end if;
end process;
end not2_beh;
--Program for 8x1Multiplexer in Structural Model
library IEEE;
use IEEE.std_logic_1164.all;
entity mux8x1_str is
    port ( a0: in STD_LOGIC;
           a1: in STD_LOGIC;
           a2: in STD_LOGIC;
           a3: in STD_LOGIC;
           a4: in STD_LOGIC;
           a5: in STD_LOGIC;
           a6: in STD_LOGIC;
           a7: in STD_LOGIC;
           s : in STD_LOGIC_VECTOR(2 downto 0);
           f: out STD_LOGIC);
end mux8x1_str;

architecture mux8x1_str of mux8x1_str is
    component and4_dat is
        port(a,b,c,d: in STD_LOGIC;
              p: out STD_LOGIC);
    end component;
    component or8_dat is
        port(a,b,c,d,e,f,g,h: in STD_LOGIC;
              p: out STD_LOGIC);
    end component;
    component not2_beh is
        port(a: in STD_LOGIC;
              c: out STD_LOGIC);

```

```

end component;

signal t0,t1,t2,t3,t4,t5,t6,t7,t8,t9,t10: STD_LOGIC;
begin
n1: not2 port map(s(0),t0);
n2: not2 port map(s(1),t1);
n3: not2 port map(s(2),t2);
d1: and4_dat port map(a0,t0,t1,t2,t3);
d2: and4_dat port map(a1,s(0),t1,t2,t4);
d3: and4_dat port map(a2,t0,s(1),t2,t5);
d4: and4_dat port map(a3,s(0),s(1),t2,t6);
d5: and4_dat port map(a4,t0,t1,s(2),t7);
d6: and4_dat port map(a5,s(0),t1,s(2),t8);
d7: and4_dat port map(a6,t0,s(1),s(2),t9);
d8: and4_dat port map(a7,s(0),s(1),s(2),t10);
r1: or8_dat port map(t3,t4,t5,t6,t7,t8,t9,t10,f);
end mux8x1_str;

```

TESTBENCH:

```

LIBRARY ieee;
USE ieee.std_logic_1164.ALL;
USE ieee.std_logic_unsigned.all;
USE ieee.numeric_std.ALL;

ENTITY tb_kpmux_vhd IS
END tb_kpmux_vhd;
ARCHITECTURE behavior OF tb_kpmux_vhd IS
-- Component Declaration for the Unit Under Test (UUT)
COMPONENT kpmux8x1
PORT(
    s : IN std_logic_vector(2 downto 0);
    d : IN std_logic_vector(7 downto 0);
    y : OUT std_logic
);
END COMPONENT;

--Inputs
SIGNAL s : std_logic_vector(2 downto 0) := (others=>'0');
SIGNAL d : std_logic_vector(7 downto 0) := (others=>'0');

--Outputs
SIGNAL y : std_logic;
BEGIN
-- Instantiate the Unit Under Test (UUT)

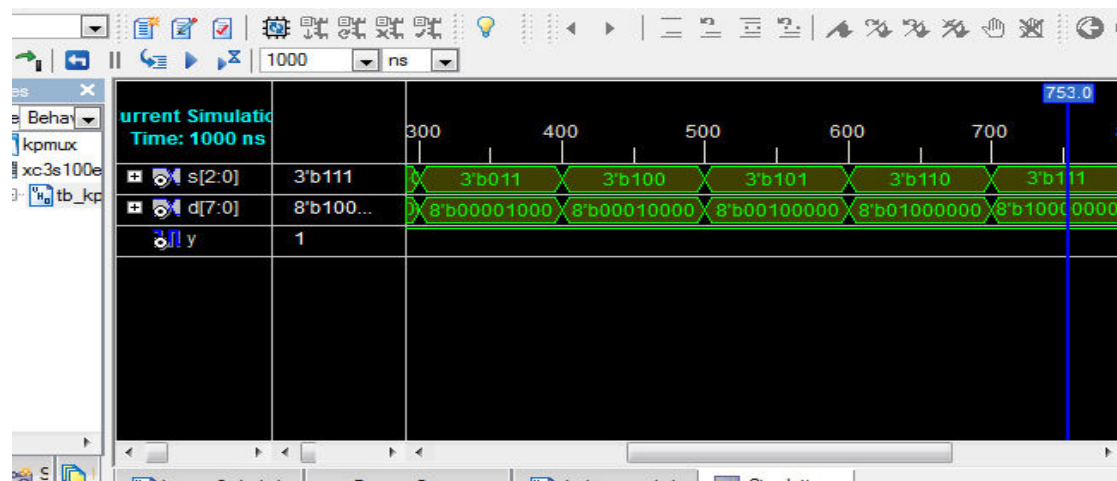
```

```

uut: kpmux8x1 PORT MAP(
    s => s,
    d => d,
    y => y
);
tb : PROCESS
BEGIN
    -- Wait 100 ns for global reset to finish
    s<= "000"; d<="00000001";
    wait for 100 ns;
    s<= "001"; d<="00000010";
    wait for 100 ns;
    s<= "010"; d<="00000100";
    wait for 100 ns;
    s<= "011"; d<="00001000";
    wait for 100 ns;
    s<= "100"; d<="00010000";
    wait for 100 ns;
    s<= "101"; d<="00100000";
    wait for 100 ns;
    s<= "110"; d<="01000000";
    wait for 100 ns;
    s<= "111"; d<="10000000";
    -- Place stimulus here
    wait; -- will wait forever
END PROCESS;
END;

```

WAVEFORMS:



RESULT: All models of 8x1 Multiplexer are simulated and waveforms are observed.

4.2 2x4 DEMULTIPLEXER

Exp No:5

AIM: Simulate the programs of 2x4 Demultiplexer using VHDL in Data flow, Behavioral and Structural models.

APPARATUS REQUIRED:

XILINX 9.2I /Active HDL4.2 software.

PROGRAM:

Data flow Model

Code:

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx primitives in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity demux2x4_dat is
    port ( a: in STD_LOGIC;
           s0: in STD_LOGIC;
           s1: in STD_LOGIC;
           y0: out STD_LOGIC;
           y1: out STD_LOGIC;
           y2: out STD_LOGIC;
           y3: out STD_LOGIC);
end demux2x4_dat;
architecture demux2x4_dat of demux2x4_dat is
begin
    y0<=(not s0) and (not s1) and a;
    y1<= s0 and (not s1) and a;
    y2<=(not s0) and s1 and a;
    y3<= s0 and s1 and a;
end demux2x4_dat;
```

Test bench:

```
-----
-- Company:
-- Engineer:
--
-- Create Date: 10:39:18 01/05/2022
-- Design Name:
-- Module Name: C:/WINDOWS/system32/kp/tb_demux2x4.vhd
```

```

-- Project Name: kp
-- Target Device:
-- Tool versions:
-- Description:
--
-- VHDL Test Bench Created by ISE for module: demux2x4_dat
--
-- Dependencies:
--
-- Revision:
-- Revision 0.01 - File Created
-- Additional Comments:
--
-- Notes:
-- This testbench has been automatically generated using types std_logic and
-- std_logic_vector for the ports of the unit under test.  Xilinx recommends
-- that these types always be used for the top-level I/O of a design in order
-- to guarantee that the testbench will bind correctly to the post-implementation
-- simulation model.

```

```

-----
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;

```

```

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--USE ieee.numeric_std.ALL;

```

```

ENTITY tb_demux2x4 IS
END tb_demux2x4;

```

```

ARCHITECTURE behavior OF tb_demux2x4 IS

```

```

    -- Component Declaration for the Unit Under Test (UUT)

```

```

    COMPONENT demux2x4_dat
    PORT(
        a : IN std_logic;
        s0 : IN std_logic;
        s1 : IN std_logic;
        y0 : OUT std_logic;
        y1 : OUT std_logic;
        y2 : OUT std_logic;
        y3 : OUT std_logic
    );

```

```

    END COMPONENT;

```

```

    --Inputs

```

```

    signal a : std_logic := '0';
    signal s0 : std_logic := '0';
    signal s1 : std_logic := '0';

```

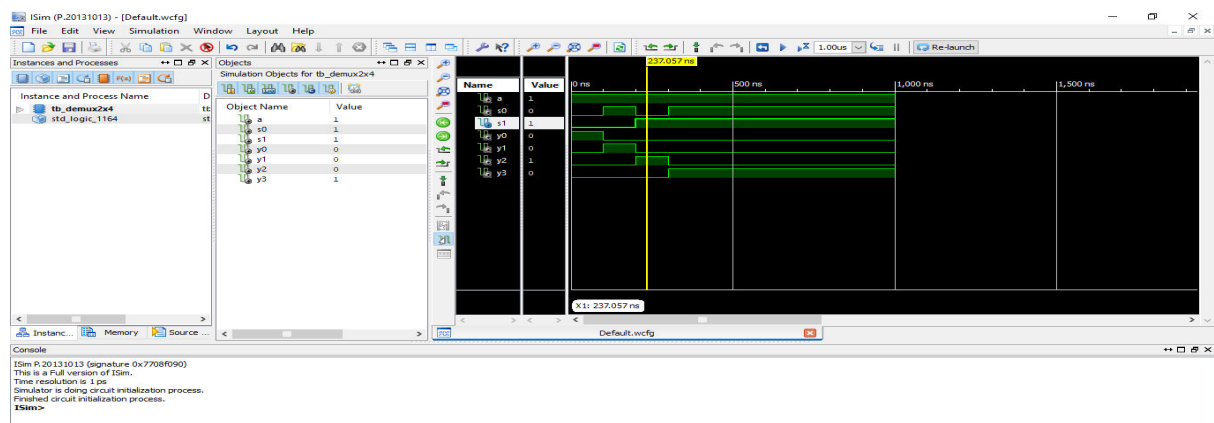
```

--Outputs
signal y0 : std_logic;
signal y1 : std_logic;
signal y2 : std_logic;
signal y3 : std_logic;
-- No clocks detected in port list. Replace <clock> below with
-- appropriate port name
-- constant <clock>_period : time := 10 ns;
BEGIN
    -- Instantiate the Unit Under Test (UUT)
    uut: demux2x4_dat PORT MAP (
        a => a,
        s0 => s0,
        s1 => s1,
        y0 => y0,
        y1 => y1,
        y2 => y2,
        y3 => y3
    );

    -- Clock process definitions
    -- Stimulus process
    stim_proc: process
    begin
        -- hold reset state for 100 ns.
        s1<='0';s0<='0';a<='1';
        wait for 100 ns;
        s1<='0';s0<='1';a<='1';
        wait for 100 ns;
        s1<='1';s0<='0';a<='1';
        wait for 100 ns;

        -- insert stimulus here
        s1<='1';s0<='1';a<='1';
        wait;
    end process;
END;

```



RESULT: All models of 2x4 Demultiplexer are simulated and waveforms are observed.

5. D- FLIP-FLOP

Exp No: 6

AIM: Simulate the program of D Flop-flop using VHDL in Behavioral model.

APPARATUS REQUIRED:

XILINX 9.2I /Active HDL4.2 software.

PROGRAM:

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx primitives in this code.
--library UNISIM;
--use UNISIM.VComponents.all;
entity dff7474 is
  Port ( clk1, clk2, clr_11, clr_12, pr_11, pr_12, d1, d2 : in STD_LOGIC;
        q1,qbar1, q2, qbar2 : out STD_LOGIC);
end dff7474;
architecture Behavioral of dff7474 is
begin
  p1: process (clk1, clr_11, pr_11, d1)
  begin
    if (pr_11 = '0' and clr_11 = '0') then q1<= '0'; qbar1 <= '1';
    elsif (pr_11 = '0') then q1<= '1'; qbar1 <= '0';
    elsif (clr_11 = '0') then q1<= '0'; qbar1 <= '1';
    elsif (pr_11 = '1' and clr_11 = '1' and (clk1'event and clk1 = '1')) then q1<= d1; qbar1 <=
    not d1;
    end if;
  end process;
  p2: process (clk2, clr_12, pr_12, d2)
  begin
    if (pr_12 = '0' and clr_12 = '0') then q2<= '0'; qbar2 <= '1';
    elsif (pr_12 = '0') then q2<= '1'; qbar2 <= '0';
    elsif (clr_12 = '0') then q2<= '0'; qbar2 <= '1';
    elsif (pr_12 = '1' and clr_12 = '1' and (clk2'event and clk2 = '1')) then q2<= d2; qbar2 <=
    not d2;
    end if;
  end process;
end Behavioral;
Test bench:
-----
-- Company:
```

```

-- Engineer:
-- Create Date: 11:15:49 01/05/2022
-- Design Name:
-- Module Name: C:/WINDOWS/system32/kp/tb_dff7474.vhd
-- Project Name: kp
-- Target Device:
-- Tool versions:
-- Description:
--
-- VHDL Test Bench Created by ISE for module: dff7474
--
-- Dependencies:
--
-- Revision:
-- Revision 0.01 - File Created
-- Additional Comments:
--
-- Notes:
-- This testbench has been automatically generated using types std_logic and
-- std_logic_vector for the ports of the unit under test.  Xilinx recommends
-- that these types always be used for the top-level I/O of a design in order
-- to guarantee that the testbench will bind correctly to the post-implementation
-- simulation model.

```

```

-----
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;

```

```

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--USE ieee.numeric_std.ALL;

```

```

ENTITY tb_dff7474 IS
END tb_dff7474;

```

```

ARCHITECTURE behavior OF tb_dff7474 IS

```

```

    -- Component Declaration for the Unit Under Test (UUT)
    COMPONENT dff7474
    PORT(
        clk1 : IN std_logic;
        clk2 : IN std_logic;
        clr_l1 : IN std_logic;
        clr_l2 : IN std_logic;
        pr_l1 : IN std_logic;
        pr_l2 : IN std_logic;
        d1 : IN std_logic;
        d2 : IN std_logic;
        q1 : OUT std_logic;
        qbar1 : OUT std_logic;
        q2 : OUT std_logic;
    );

```

```

        qbar2 : OUT std_logic
    );
END COMPONENT;
--Inputs
signal clk1 : std_logic := '0';
signal clk2 : std_logic := '0';
signal clr_11 : std_logic := '0';
signal clr_12 : std_logic := '0';
signal pr_11 : std_logic := '0';
signal pr_12 : std_logic := '0';
signal d1 : std_logic := '0';
signal d2 : std_logic := '0';

--Outputs
signal q1 : std_logic;
signal qbar1 : std_logic;
signal q2 : std_logic;
signal qbar2 : std_logic;

-- Clock period definitions
-- constant clk1_period : time := 10 ns;
-- constant clk2_period : time := 10 ns;

BEGIN

    -- Instantiate the Unit Under Test (UUT)
    uut: dff7474 PORT MAP (
        clk1 => clk1,
        clk2 => clk2,
        clr_11 => clr_11,
        clr_12 => clr_12,
        pr_11 => pr_11,
        pr_12 => pr_12,
        d1 => d1,
        d2 => d2,
        q1 => q1,
        qbar1 => qbar1,
        q2 => q2,
        qbar2 => qbar2
    );

    -- Stimulus process
    stim_proc: process
    begin
        -- hold reset state for 100 ns.
        pr_11<='0';clr_11<='0';
            wait for 100 ns;

        pr_11<='0';
        wait for 100 ns;
        clr_11<='0';

```

```

        wait for 100 ns;
        pr_l1<='1';clr_l1<='1';clk1<='1';d1<='1';
        wait for 100 ns;
-- wait for clk1_period*10;
        pr_l2<='0';clr_l2<='0';
        wait for 100 ns;

        pr_l2<='0';
        wait for 100 ns;
        clr_l2<='0';

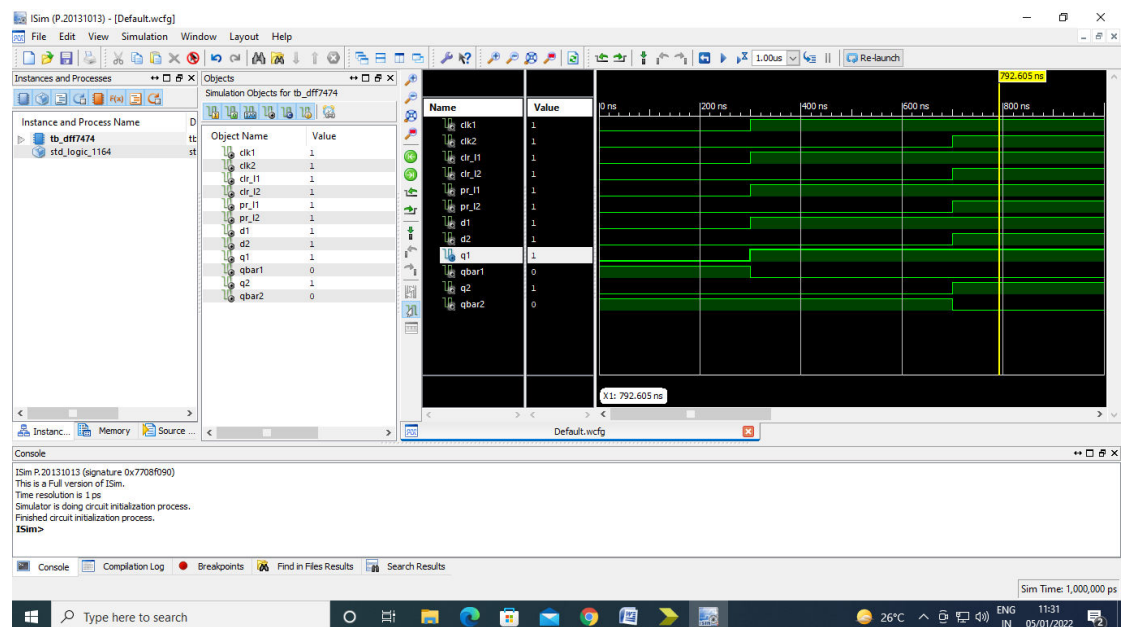
        wait for 100 ns;
        pr_l2<='1';clr_l2<='1';clk2<='1';d2<='1';
        -- insert stimulus here

        wait;
    end process;

END;

```

WAVEFORMS:



RESULT: The Behavioral model of D Flip-Flop are simulated and waveforms are observed.

6.SHIFT REGISTER

Date:

Exp No: 8

AIM: Simulate the program of Shift register using VHDL in Behavioral model.

APPARATUS REQUIRED:

XILINX 9.2I /Active HDL4.2 software.

PROGRAM:

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;

entity Shift_register_VHDL is

    port( clk: in std_logic;
          L: in std_logic;
          w: in std_logic; -- new data to shift in
          Output: out std_logic_vector(3 downto 0);
          Input: in std_logic_vector( 3 downto 0));

end Shift_register_VHDL;

architecture Behavioral of Shift_register_VHDL is
begin
    process
        variable temp: std_logic_vector(3 downto 0);
    begin
        wait until rising_edge (clk);
        temp := Input;
        if L='1' then
            for i in 0 to 2 loop
                temp(i) := temp(i+1);
            end loop;
            temp(3) := w;
        end if;
        Output <= temp;
    end process;
end Behavioral;
```

TEST BENCH:

```
-----
-- Company:
-- Engineer:
--
-- Create Date: 16:49:42 01/06/2022
-- Design Name:
```



```

-- Module Name: C:/WINDOWS/system32/kp33/tb_shift.vhd
-- Project Name: kp33
-- Target Device:
-- Tool versions:
-- Description:
--
-- VHDL Test Bench Created by ISE for module: Shift_register_VHDL
--
-- Dependencies:
--
-- Revision:
-- Revision 0.01 - File Created
-- Additional Comments:
--
-- Notes:
-- This testbench has been automatically generated using types std_logic and
-- std_logic_vector for the ports of the unit under test. Xilinx recommends
-- that these types always be used for the top-level I/O of a design in order
-- to guarantee that the testbench will bind correctly to the post-implementation
-- simulation model.

```

```

-----
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--USE ieee.numeric_std.ALL;
ENTITY tb_shift IS
END tb_shift;
ARCHITECTURE behavior OF tb_shift IS
    -- Component Declaration for the Unit Under Test (UUT)
    COMPONENT Shift_register_VHDL
    PORT(
        clk : IN std_logic;
        L : IN std_logic;
        w : IN std_logic;
        Output : OUT std_logic_vector(3 downto 0);
        Input : IN std_logic_vector(3 downto 0)
    );
    END COMPONENT;
--Inputs
signal clk : std_logic := '0';
signal L : std_logic := '0';
signal w : std_logic := '0';
signal Input : std_logic_vector(3 downto 0) := (others => '0');

--Outputs
signal Output : std_logic_vector(3 downto 0);

-- Clock period definitions

```

```

--constant clk_period : time := 10 ns;
BEGIN
    -- Instantiate the Unit Under Test (UUT)
    uut: Shift_register_VHDL PORT MAP (
        clk => clk,
        L => L,
        w => w,
        Output => Output,
        Input => Input
    );
    -- Stimulus process
    stim_proc: process
    begin
        -- hold reset state for 100 ns.
        clk<='1'; L<='1';w<='1'; Input<="0101";
        wait for 100 ns;

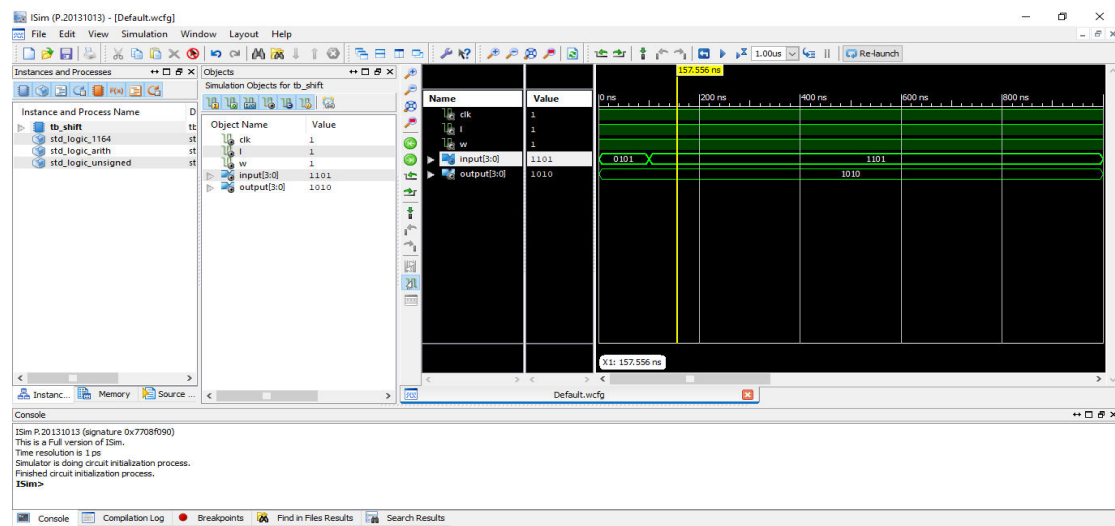
        --wait for clk_period*10;

        -- insert stimulus here
        clk<='1'; L<='1';w<='1'; Input<="1101";
        wait;
    end process;

```

END;

WAVEFORMS:



RESULT: The Behavioral model of Shift register is simulated and waveforms are observed.

DECADE COUNTER:

Date:

Exp No: 7

AIM: Simulate the program of Decade counter using VHDL in Behavioral model.

APPRATUS REQUIRED:

Active HDL4.2 software.

PROGRAM:

```
library ieee;
use ieee.std_logic_1164.all;
use ieee.std_logic_unsigned.all;
use ieee.std_logic_arith.all;
entity counter7 is
    port(clk:in std_logic;
         rst:in std_logic;
         tc:out std_logic;
         count:buffer std_logic_vector(3 downto 0));
end counter7;
architecture counter71 of counter7 is
begin
    process(clk,rst)
begin
    if(clk'event and clk='1')then
    if(rst='1'or count="1001")then
        count<="0000";
    else
        count<=count+1;
    end if;
    end if;
    if(count="1001")then
        tc<='1';
    else
        tc<='0';
    end if;
    end process;
end counter71;
```

WAVEFORMS:

RESULT: The Behavioral model of Decade counter are simulated and waveforms are observed.